Tests will be graded & returned by Friday 9-25

Tests will be graded by Friday.

Some students are struggling to show they know concepts.

Therefore I will post grades, but not return exams. Grades will be curved.

Optional: Re-do the (same) test as a take-home exam

If the take-home score > in-class score, then

the two scores will be averaged for the test 1 grade.

The curve will be the same as for the in-class exam.

Ch5 S7 continued

canonical form of

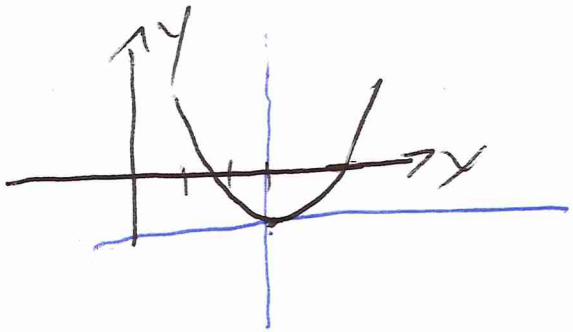
parabola $y = x^2$

circle $x^2 + y^2 = 1$

hyperbola $x^2 - y^2 = 1$

or $xy = 1$

$y = (x-3)^2 + 1$



canonical forms of 2nd-order linear PDE's $au_{\xi\xi} + bu_{\xi\eta} + cu_{\eta\eta} + \ldots = 0$

$\Delta = b^2 - 4ac$

hyperbolic $(\Delta > 0)$ $u_{\xi\xi} - u_{\eta\eta} = 0$ → more physical

$u_{\xi\eta} = 0$ → easy to solve mathematically

elliptic $(\Delta < 0)$ $u_{\xi\xi} + u_{\eta\eta} = 0$

parabolic $(\Delta = 0)$ $u_{\xi\xi} + \ldots = 0$

<u>Thm</u> If a 2nd order linear PDE is hyp / parab / elliptic in a domain Ω, Then at a pt $x_0 \in \Omega$, we can find a coord system (ξ, η) so that the eq'n has the corresponding canonical form.

Note: unless the coeff are constant, the coord system (ξ, η) might be different at every pt in Ω.

Change independent variables

$u(x,y) \to u(\xi,\eta)$ given $\xi = \xi(x,y)$
$\eta = \eta(x,y)$

Check: is transformation ok?

$$J = \begin{vmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \xi}{\partial y} \\ \frac{\partial \eta}{\partial x} & \frac{\partial \eta}{\partial y} \end{vmatrix} \neq 0, \pm\infty.$$

ex) $\frac{\partial u}{\partial x} - 2\frac{\partial u}{\partial y} + 3yu = e^x$

$\xi = x + 2y \qquad \eta = \frac{1}{2}x + 2y$

$$J = \begin{vmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \xi}{\partial y} \\ \frac{\partial \eta}{\partial x} & \frac{\partial \eta}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ \frac{1}{2} & 2 \end{vmatrix} = 2 - 1 = 1 \neq 0$$

good

start with total differential

$$\frac{\partial}{\partial x}\Big|_y \left[du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = \underbrace{\frac{\partial u}{\partial \xi}d\xi + \frac{\partial u}{\partial \eta}d\eta}_{\text{start here because we want to end with } u_\xi,\, u_\eta} \right]$$

$$\left.\frac{\partial u}{\partial x}\right|_y = \frac{\partial u}{\partial \xi}\left.\frac{\partial \xi}{\partial x}\right|_y + \frac{\partial u}{\partial \eta}\left.\frac{\partial \eta}{\partial x}\right|_y$$

$$= u_\xi (1) + \frac{\partial u}{\partial \eta}\left(\frac{1}{2}\right)$$

$$\left.\frac{\partial u}{\partial y}\right|_x = \frac{\partial u}{\partial \xi}\left.\frac{\partial \xi}{\partial y}\right|_x + \frac{\partial u}{\partial \eta}\left.\frac{\partial \eta}{\partial y}\right|_x$$

$$= u_\xi (2) + u_\eta (2)$$

$$\Delta u = \frac{\partial u}{\partial \xi}\Delta \xi + \frac{\partial u}{\partial \eta}\Delta \eta + \cdots (\Delta \xi)^2 \quad (\Delta \eta)^2$$

$$\lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x} = \frac{\partial u}{\partial \xi}\frac{\Delta \xi}{\Delta x} + \frac{\partial u}{\partial \eta}\frac{\Delta \eta}{\Delta x}$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x} + \frac{\partial u}{\partial \eta}\cancel{\lim_{\Delta x \to 0}\frac{\Delta y}{\Delta x}} + \cdots \frac{\partial \eta}{\partial x}$$

subst into PDE:

$$u_\xi + \frac{1}{2}u_\eta - 2\left[2u_\xi + 2u_\eta\right] + 3\left(-\frac{1}{2}\xi + \eta\right)u = e^{2\xi - 2\eta}$$

we need $x(\xi,\eta)$ $y(\xi,\eta)$

$$\begin{bmatrix} \xi \\ \eta \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ \frac{1}{2} & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{1}\begin{bmatrix} 2 & -2 \\ -\frac{1}{2} & 1 \end{bmatrix}\begin{bmatrix} \xi \\ \eta \end{bmatrix}$$

$$x = 2\xi - 2\eta$$
$$y = -\frac{1}{2}\xi + \eta$$

ex) $u_x + u_y = 0$ convert to polar coord.

$x = r\cos\theta \quad y = r\sin\theta$

$r^2 = x^2 + y^2$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= r\cos^2\theta + r\sin^2\theta$$

$$= r(\cos^2\theta + \sin^2\theta) = r$$

$$du = \cancel{\frac{\partial u}{\partial \xi} d\xi + \frac{\partial u}{\partial \eta} d\eta}$$

$$\frac{\partial}{\partial x}\Big|_y \left[du = \frac{\partial u}{\partial r} dr + \frac{\partial u}{\partial \theta} d\theta \right]$$

$$\frac{\partial u}{\partial x}\Big|_y = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x}\Big|_y + \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial x}\Big|_y$$

$$= u_r[\cos\theta] + u_\theta\left[\frac{-\sin\theta}{r}\right]$$

$\frac{\partial r}{\partial x}\Big|_y$: $r^2 = x^2 + y^2$

$2r\frac{\partial r}{\partial x}\Big|_y = 2x$

$r_x = \frac{x}{r}$ ← don't want x!

$= \frac{r\cos\theta}{r} = \cos\theta$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$y = r\sin\theta$
$x = r\cos\theta$

$$\frac{\partial \theta}{\partial y}\Big|_x = \frac{1}{1+\left(\frac{y}{x}\right)^2}\left[\frac{1}{x}\right] = \dots = \frac{\cos\theta}{r}$$

$\frac{\partial r}{\partial y}\Big|_x$: $r^2 = x^2 + y^2$

$2r\frac{\partial r}{\partial y} = 2y$

$r_y = \frac{y}{r} = \frac{r\sin\theta}{r} = \sin\theta$

$$\left.\frac{\partial u}{\partial y}\right|_x = \frac{\partial u}{\partial r}\left.\frac{\partial r}{\partial y}\right|_x + \frac{\partial u}{\partial \theta}\left.\frac{\partial \theta}{\partial y}\right|_x$$

$$= u_r \sin\theta + u_\theta \frac{\cos\theta}{r}$$

$$\underline{(\cos\theta + \sin\theta)\, u_r + \frac{1}{r}(\cos\theta - \sin\theta)\, u_\theta = 0}$$